1.5 Towards The Twentieth Century

1878

Out of the blue (Cuba, to be precise), a Spaniard called Ramón Verea came to New York City at the end of the US Civil War. He traded Spanish gold and banknotes, which got him interested in calculation. Necessity is... and so, this year, Verea created a calculating machine and was granted a patent for it.

There were calculators before Verea's invention, but they all performed repeated addition operations to get a multiplication result. Verea's contribution lay in seeing how to perform multiplication with one stroke of a lever. His machine was based on a ten-sided metal cylindrical structure.

Scientific American ran an article about Verea's calculator, but the man never tried to market his own invention—almost in scorn. He seemed to have just wanted to prove that "backward" Spaniards could come up with inventions just like Americans and Englishmen could.

By the end of the century, all mechanical-calculation systems had switched over to Verea-type systems.

1885

One more mass-production milestone: American Frank Baldwin and T Odhner, a Swede living in Russia, independently and simultaneously came up with a multiplying calculator that was more compact than the Arithmometer. This was not based on, but was a significant extension of, Leibniz's machine (see 1671). Again, space constraints dictate that we direct you to a URL to find out more: **www.xnumber.com/xnumber/mechanical2.htm**.

1886 And 1889

In 1886, Dorr Felt, who used to work in a machine shop, created, in Chicago, his Comptometer. This was the first mechanical calculator where the data was entered by pressing keys rather than by